

Landsat and Sentinel-2 Based Flood Mapping of the September 2024 Central European Flood Event

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1 General Information

Topic: Landsat and Sentinel-2 Based Flood Mapping of the September 2024 Central European Flood Event

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Source Code: [Remote-Sensing Project on GitHub](#)
Dataset: [Landsat Dataset with Flood Events](#);
[Sentinel Dataset with Flood Events](#)

2 Abstract

This project aims to develop a machine learning-based approach for assessing flooded areas using satellite imagery, with a focus on the Central European flood event of September 2024, primarily caused by Storm Boris. The event led to significant flooding across Austria, Poland, Slovakia, and Germany, resulting in strong economic impacts. Leveraging multispectral optical imagery from the Landsat and Sentinel-2 satellite series—including RGB and near-infrared (NIR) bands — the project focuses on estimating the likelihood of water presence at the pixel level, extended by more in-depth analysis of the model performance in flooded areas.

The core of the methodology involves training a convolutional neural network for semantic segmentation (DeepLabV3+) to detect water-covered areas, extended with Monte Carlo dropout to generate uncertainty-aware predictions. This probabilistic framework enables the production of confidence maps that quantify the reliability of flood detection across spatial regions. The workflow includes data collection, preprocessing (e.g., cloud masking, tiles splitting), and feature engineering (e.g., NDWI), followed by model training and evaluation using standard metrics such as IoU, F1 score, Precision, Recall, and Dice Coefficient.

Ultimately, the project aims to assess the utility of optical satellite data in flood mapping and demonstrate the model's ability to distinguish floodwaters from other land surfaces with quantifiable confidence.

3 Motivation and Problem Statement

In September 2024, Central and Eastern Europe experienced one of the most devastating natural disasters in recent years when Storm Boris triggered a strong rainfall—up to 500 mm in under 48 hours—resulting in widespread flooding. The event affected countries including Austria, Czech Republic, Slovakia, Poland, Romania, Hungary, and Germany. With 27 confirmed fatalities and economic losses exceeding several billion euros, the disaster highlights the importance of effective flood monitoring tools.

Satellite-based Earth observation is a powerful tool for disaster monitoring, but accurately identifying floodwaters remains challenging. Radar sensors like Sentinel-1 can operate under cloud cover, but optical sensors (e.g., Landsat and Sentinel-2) offer better spatial and spectral resolution and are more readily available. However, their effectiveness is limited by atmospheric conditions such as cloud cover, making water classification more complex.

Traditional flood detection methods often rely on binary classification (flooded vs. non-flooded), providing no indication of prediction confidence. This restricts the usefulness of such tools in real-world emergency scenarios where uncertainty matters. This project is motivated by the need to go beyond deterministic outputs by developing a machine learning model that estimates water likelihood at the pixel level, integrating multispectral satellite data and deep learning-based segmentation with an uncertainty estimation technique - Monte Carlo dropout.

4 Research Questions

The project aims to answer the following research questions:

1. **To what extent can high-quality waterbody masks be generated using NDWI and Sentinel-2 Scene Classification Layer (SCL) data?**

Initial waterbody labels will be generated using thresholding on the NDWI index. The resulting masks will be evaluated through manual visual inspection (Sentinel-based part of the dataset) and quantitative comparison with reference datasets, e.g., JRC Global Surface Water (Landsat-based part of the dataset). The primary evaluation metric will be the Correction Rate, defined as the proportion of masks that required manual refinement after automatic generation. This assessment aims to determine whether the semi-automated labeling approach produces annotations of sufficient quality for model training.

2. **How accurately can the proposed deep learning model identify water-covered areas in satellite imagery?**

This question will be addressed by evaluating the final trained model on the labelled test dataset. The model's performance in detecting water-covered areas will be measured using standard image segmentation metrics, including precision, recall, F1 score, and Intersection over Union (IoU). These metrics will quantify the model's effectiveness in identifying waterbodies and flooded regions.

3. **To what extent can a model trained on Sentinel-2 data be transferred to Landsat imagery?**

To assess cross-sensor generalization, the model trained on Sentinel-2 data will be directly applied to Landsat images. Performance will be evaluated using the same segmentation metrics (IoU, F1 score, precision, recall), and results will be compared against those obtained on Sentinel-2 test data. A significant drop in performance will indicate limited transferability.

4. **Can uncertainty estimates be used to improve the reliability of flood classification?**

The analysis will focus specifically on images captured during flood events. Uncertainty will be estimated using Monte Carlo Dropout. At multiple uncertainty thresholds, low-confidence pixels will be filtered out, and segmentation quality will be re-evaluated on the remaining, high-confidence areas. Metrics such as precision, recall, F1 score, and Intersection over Union (IoU) will be computed for multiple thresholds.

5 Methodology

This project follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology to ensure a structured, iterative, and reproducible approach. The main phases include domain understanding, data preparation, modeling, evaluation, and documentation of results.

5.1 Business Understanding

The project began with a review of the September 2024 Central European flood event, analyzing its scope and affected regions. In addition, relevant studies [2], [4], [5], were reviewed to understand how similar research is structured and conducted.

5.2 Data Understanding

The next step involved a detailed assessment of the data sources required for effective flood detection. Multispectral imagery from Sentinel-2 and Landsat-8/9 was selected due to its high spatial resolution, frequent revisit rates, and broad geographic coverage. Several challenges were identified early on, including the presence of cloud cover in optical imagery, spectral confusion between water, snow, and urban surfaces, and the need for accurate, event-specific ground truth labels to support supervised learning.

5.3 Data Preparation

5.3.1 Data Extraction

The first stage of this project involved building a custom satellite imagery dataset. Images from both satellite series **Sentinel-2** and **Landsat** were considered during the project. Landsat, operated by NASA and the USGS, provides a long-term archive of Earth observation data with moderate spatial resolution (30 meters) and a 16-day revisit cycle, making it ideal for historical flood analysis. In contrast, Sentinel-2, part of the Copernicus Programme by the European Space Agency (ESA), offers higher spatial resolution (10–20 meters) and more frequent revisit intervals (every five days using Sentinel-2A and 2B), which is critical for timely flood monitoring and response.

To construct a representative and high-quality dataset, a data extraction pipeline using the Sentinel Hub platform was implemented. A stratified random sampling strategy was applied to ensure that a wide range of climatic, atmospheric, and land-cover conditions were captured. Data acquisition spanned the years 2022–2024, with equal representation from all four quarters of each year. A maximum cloud cover threshold of 50% was enforced to filter out heavily obscured images in the Sentinel dataset. For Landsat images with more than 40% cloud cover or over 10% invalid pixels were automatically filtered out based on mask generation described below.

The Sentinel dataset was collected in three main steps:

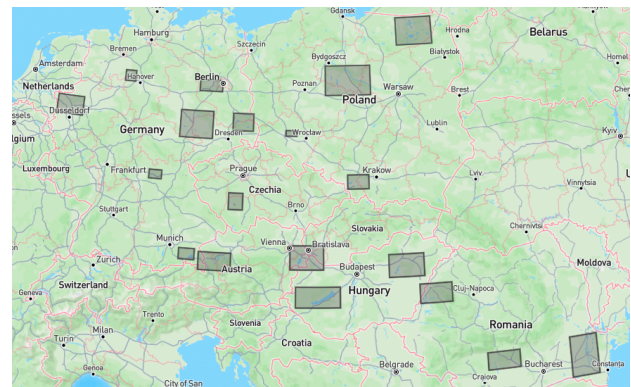
- 1. General Sampling:** Geojson polygon was defined (see Figure 1a) encompassing the countries of interest and randomly selected points within these areas. From each selected point, 2048×2048 pixel tiles that met the cloud cover condition were extracted. This process was repeated quarterly for each year, with 50 images per quarter collected to ensure seasonal balance.
- 2. Waterbody-Enriched Sampling:** To increase the proportion of images containing visible water bodies, regions with high densities of lakes and rivers were identified and targeted (see Figure 1b). Additional samples were extracted from these waterbody-rich areas for each quarter.

3. Event-Specific Sampling: To capture flooding events explicitly, imagery taken after the following documented floods (the time period specified for each data extraction step is indicated in parentheses) were included in the dataset:

- Storm Boris (2024-09-09 - 2024-09-30)
- Bavaria Flood (2024-05-30 - 2024-06-08)
- Northern Germany Flood (2021-07-14 - 2021-08-05)



(a) Areas of Interest for general sampling.



(b) Areas of Interest for waterbodies sampling.

Figure 1: Regions selected for satellite data sampling.

The Landsat dataset was also collected in three main steps:

- 1. General area sampling:** For each quarter of 2022, 2023, and 2024, 40 images of size 684×684 pixels (30 m spatial resolution) were downloaded. Each image was centred on a random point within the broader area of interest encompassing flood-prone regions.
- 2. Water-rich regions:** An additional set of 10 images per quarter was collected from predefined smaller areas known for dense waterbody presence, based on a GeoJSON file.
- 3. Flood event sampling:** For each of the three chosen flood events, 10 images were selected over a 20-day period before, during, and after the event.

For each **Sentinel-2** scene, the following bands were extracted: B02, B03, B04, B08 (Blue, Green, Red, and Near Infrared), CLD (Cloud Mask), dataMask (Valid Pixel Mask), SCL (Scene Classification Layer).

For **Landsat** imagery, the extracted bands included: B02, B03, B04, B05, B06, B10 (Blue, Green, Red, NIR, SWIR-1, and Thermal), dataMask (Valid Pixel Mask).

The dataset extraction code can be found in the corresponding Python notebooks.¹

¹[00_creating_Landsat_dataset.ipynb](#) and [00_creating_Sentinel_dataset.ipynb](#)

5.3.2 Mask Annotation and Label Generation

The next stage involved annotating the dataset by generating masks to label waterbodies and related surface features. To generate masks for water and flood detection, a multi-step pipeline that combines spectral index computation, scene classification layers, and cloud probability data was developed.

a. Sentinel

For the Sentinel-2 dataset, each mask encodes four distinct classes: 0 – Background, 1 – Water, 2 – Cloud, 3 – Snow/Ice. To generate these masks, combined threshold-based methods were applied to three complementary sources: NDWI, the Scene Classification Layer (SCL), and the Cloud Probability Layer (CLD).

Cloud mask: cloud detection is performed using the CLD band. A pixel is marked as cloudy if its cloud probability exceeds a threshold of 20%. Cloudy pixels are overlaid in red and override any water or ice classification.

Water mask: water masks were created as a combination of SCL-water band and the Normalized Difference Water Index (NDWI). NDWI is computed as:

$$\text{NDWI} = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR} + \epsilon}$$

where $\epsilon = 1 \times 10^{-10}$ is a small value to avoid division by zero. Based on the NDWI values, each pixel is classified into three categories:

- $\text{NDWI} < 0.1$ – Dry or non-water areas
- $0.1 \leq \text{NDWI} < 0.2$ – Possible flooding or moisture
- $\text{NDWI} \geq 0.2$ – Water surfaces

Subsequently, the masks were iteratively refined through manual inspection and correction with the help of the `napari` Python library. This process ensured high-quality labels suitable for training and evaluation (see Figure 2 for an example of mask improvement). This manual refinement process was also used to assess the feasibility of semi-automated waterbody label generation — specifically addressing the research question Q1: *To what extent can high-quality waterbody masks be generated using NDWI and Sentinel-2 Scene Classification Layer (SCL) data?* For the Sentinel-2 part of the dataset, approximately 24% of images required manual corrections. This yields a Correction Rate of 24%, indicating that around three-quarters of the auto-generated masks were of sufficient quality for direct use in training. These findings suggest that NDWI and SCL-based methods offer a viable starting point for semi-automated label generation in waterbody segmentation tasks, significantly reducing manual annotation effort.

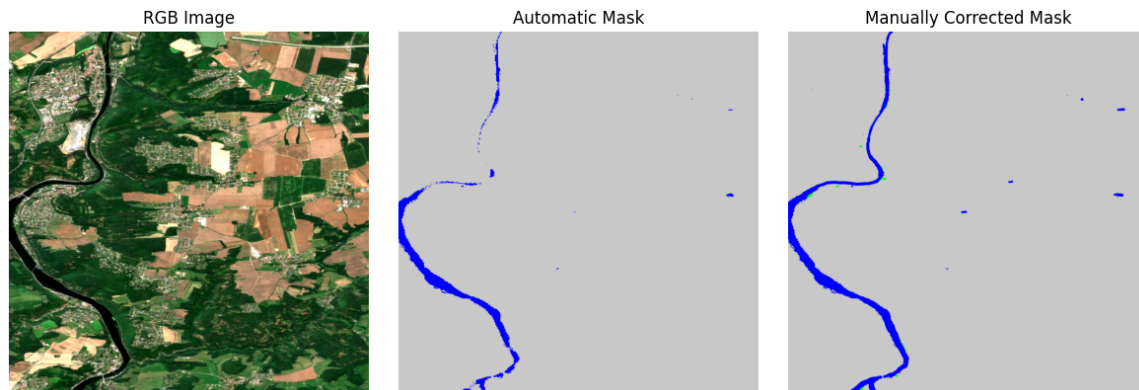


Figure 2: Comparison of RGB image, automatic mask and manually corrected mask.

The masks creation and correction code can be found in `01_correcting_masks.ipynb` Python notebook.

After creating the masks, the original 2048×2048 pixel Sentinel-2 images and their corresponding masks were split into smaller, fixed-size tiles. A tiling function that divides each image and its associated label mask into non-overlapping patches of size 256×256 pixels was implemented. This results in a 8×8 grid, yielding a total of 64 tiles per image. All pixels in the tile must be marked as valid in the `data_mask` layer. Tiles containing more than 51% cloud-covered pixels (based on red intensity in the RGB mask) were excluded to avoid noisy training examples.

For each accepted tile, two files were saved:

1. **Image Tile:** A GeoTIFF file containing four spectral bands (Red, Green, Blue, and NIR) extracted from the original `rgb_nir` image stack.
2. **Label Mask:** A corresponding mask file in which each pixel is encoded as a single class ID (background, water, cloud, or snow/ice), derived from the RGB color-coded mask.

Each tile was saved with its spatial metadata (e.g., geotransform, coordinate reference system).

The splitting code can be found in the corresponding Python notebook.²

b. Landsat

To generate semantic masks suitable for training, a custom script was implemented within Sentinel Hub, applying threshold-based classification rules derived from spectral indices. These rules relied on the Normalized Difference Snow Index (NDSI) and the Normalized Difference Vegetation Index (NDVI).

These masks were downloaded alongside the images and were also used to calculate the proportion of cloud and invalid pixels in each image. Following automatic classification, the masks were manually reviewed and refined to address snow and ice misclassifications, which often appeared as cloud due to similar brightness and temperature characteristics. A secondary postprocessing step was applied using adjusted NDSI and NDWI thresholds:

- $\text{NDSI} > 0.4$ and $\text{NDWI} < 0.3$ – Ice

²[02_splitting_Sentinel_dataset.ipynb](#)

- $\text{NDSI} < 0.4$ and $\text{NDWI} > 0.3$ – Water

```
mask[is_ice] = 0 # Reassign suspected ice to background
mask[(original == 0) & is_water] = 1 # Add water pixels if likely
```

This correction aimed to improve label consistency, though perfect separation of snow and water was not always feasible.

Furthermore, to enhance water detection, permanent water bodies from the JRC Global Surface Water dataset were used to supplement missing water pixels in the masks. These additions were made conservatively—only pixels classified as permanent water were added, with no removals performed. To ensure consistency with Sentinel-2 image dimensions and resolution, only the RGB and NIR bands were retained. All images were cropped to a 226×226 pixel window around their center and then resampled to 10 meters per pixel using the GDAL library. This transformation harmonized the spatial resolution across datasets, although the upsampled Landsat imagery naturally appeared less sharp due to the original 30 m resolution.

For training and evaluation, each transformed image was divided into non-overlapping 256×256 pixel tiles, following the same tiling and naming convention used for the Sentinel-2 dataset. This standardisation ensured seamless integration of the Landsat-derived data into the overall segmentation pipeline.

5.3.3 Final Dataset and Sampling Strategy

Since the Sentinel-2 dataset was used for training the model for this part of the project, the dataset construction and sampling strategy are discussed in greater detail here.

After the tiling process, the full dataset consisted of approximately 40,000 image–mask pairs, with each tile measuring 256×256 pixels. However, an analysis revealed that only around 2% of all pixels in the dataset were labeled as water, highlighting a strong class imbalance that could negatively impact model training.

To address this issue and improve the relevance of the dataset for water and flood segmentation tasks, an oversampling-based filtering strategy was applied. Each tile was assigned to a bin based on its water pixel percentage. The final dataset was constructed by selectively sampling from these bins as follows:

Water Pixel Percentage Bin	Number of Tiles (with oversampling)
0%	0
0–10%	5,724
10–20%	8,500
20–30%	804 (oversampled $\times 2$)
30%+	1,972 (oversampled $\times 2$)

This sampling strategy increased the proportion of water pixels to approximately 10% in the final dataset.

The final dataset was then split into three parts: training (70%), test (15%) and validation (15%). To ensure unbiased evaluation of model performance on real-world flood events, all image tiles associated with the Storm Boris flood were excluded from both training and validation. These were reserved exclusively for post-training evaluation to assess the generalisation of the model to unseen flood scenarios.

The code related to the oversampling and splitting part can be found in the corresponding Python notebook.³

5.4 Modeling

The task of segmenting waterbodies falls into the area of image segmentation, where each pixel is assigned to one of the classes and at the end, the segmented mask for the image is returned.

5.4.1 Model Architecture Discussion

After analysing articles [2], [4] that worked on waterbodies segmentation, the following architecture was selected - DeepLabV3. This architecture was selected because water can appear in varying shapes, sizes, and environmental conditions, and DeepLab is designed to handle such variability. The exact setup is **DeepLabV3** with a **ResNet-34** backbone. The model was implemented using the `segmentation_models.pytorch` library, which provides modular implementations of state-of-the-art segmentation networks. The chosen architecture, `DeepLabV3_ResNet34`, consists of the following components:

- **Encoder:** A ResNet-34 convolutional backbone pretrained on ImageNet, which extracts deep hierarchical features from the input RGB image.
- **Atrous Spatial Pyramid Pooling (ASPP):** A key feature of DeepLabV3 that captures multi-scale context and expands the model's receptive field without losing spatial resolution.
- **Decoder Head:** Produces a dense pixel-wise classification mask with four output classes: background, water, cloud, and snow/ice.

The schema of the model used for training is presented in Figure 3. The schema is based on the original paper by Chen et al. [1].

In addition to the standard segmentation output (i.e., class predictions for each pixel), the model was extended to return an uncertainty map for the water class using Monte Carlo dropout. Instead of a single softmax probability, the model performs multiple stochastic forward passes at inference time, computing the mean and standard deviation of the predicted probabilities. The mean represents the average predicted confidence for water presence, while the standard deviation serves as a measure of uncertainty. This allows the model to quantify how uncertain it is about the presence of water in each pixel—especially valuable in regions with ambiguous signals such as urban or agricultural areas that may experience temporary flooding.

5.4.2 Training Setup Discussion

The model was trained on the Sentinel-2 dataset using a configuration tailored for class-imbalanced semantic segmentation. The training setup is the following:

As an **optimizer** the AdamW was used, providing better stability and convergence for this task. The optimizer was configured with learning rate: 0.001 and weight decay: 1×10^{-4} .

³[03_dataset_extraction_jupyterhub.ipynb](#)

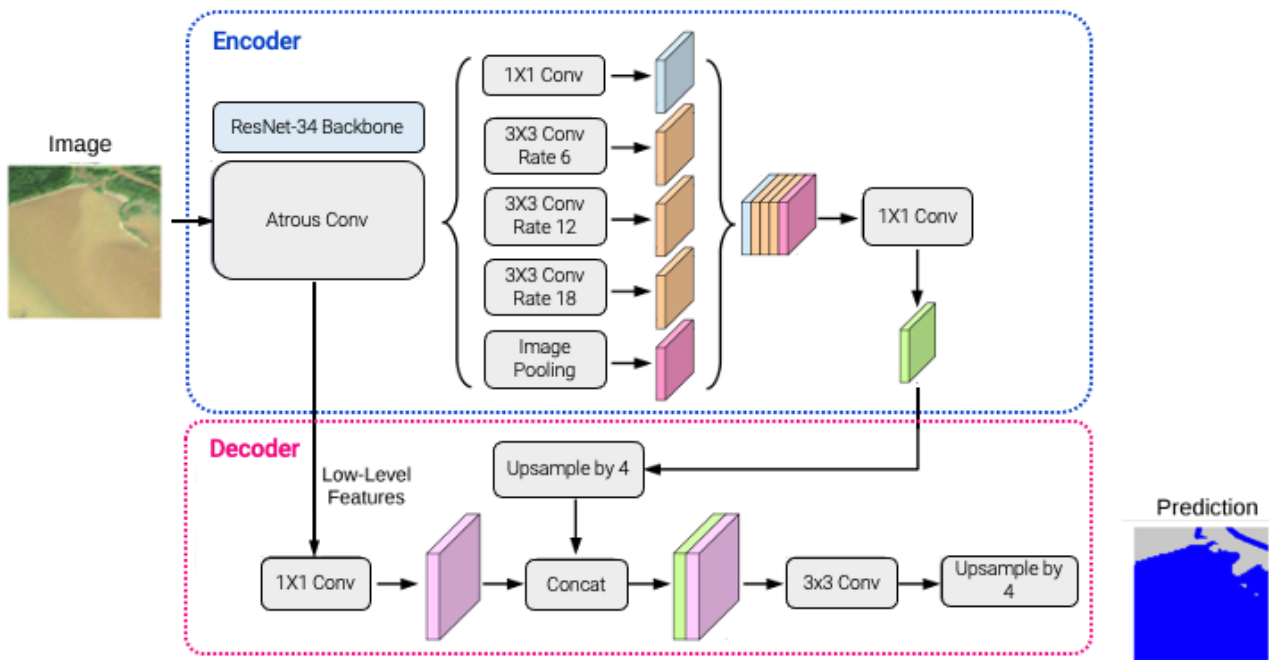


Figure 3: Architecture of the DeepLabV3+ model with ResNet34 backbone.

To dynamically adjust learning during training, a `ReduceLROnPlateau` scheduler was applied. It monitored validation performance and reduced the learning rate by a factor of 0.1 if performance plateaued for more than two consecutive evaluations. This helped prevent overfitting and allowed fine-tuning during later epochs.

Given the class imbalance—especially the underrepresentation of water pixels—a hybrid **loss function** was designed by combining:

- **Cross-Entropy Loss**, with class weights to penalize water class errors more strongly:

$$\text{class_weights} = [0.1, 0.9, 0.3, 0.3]$$

- **Dice Loss**, which focuses on spatial overlap and is particularly effective in segmentation tasks involving sparse or irregularly shaped classes.

The final loss was computed as an equal-weighted combination:

$$\mathcal{L}_{\text{final}} = 0.5 \times \mathcal{L}_{\text{CE}} + 0.5 \times \mathcal{L}_{\text{Dice}}$$

This setup balances per-pixel accuracy with region-level consistency, especially for water bodies that may appear fragmented or sparse.

The final **training configuration** is the following:

- Batch size: 128
- Number of epochs: 20
- Validation frequency: every 3 epochs
- Patience for early stopping and learning rate reduction: 3
- Subset size for initial training experiments: 8,000 samples

5.4.3 Evaluation Metrics and Experiment Tracking

To assess the performance of models both overall and class-specific metrics were tracked. The evaluation logic is encapsulated in a custom `SegmentationMetrics` class, which accumulates predictions into a confusion matrix and derives a variety of performance metrics.

For each class in the dataset (background, water, cloud, snow/ice), the following metrics were computed:

- **Precision:** Measures the proportion of predicted positives that are actually correct.
- **Recall:** Measures the proportion of actual positives that are correctly identified.
- **Dice Coefficient:** A measure of spatial overlap, between the predicted segmentation and the ground truth calculated as:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}$$

- **Intersection over Union (IoU):** Measures the overlap between the predicted segmentation and the ground truth, relative to their union, showing how well model is predicting the correct shape and position of an object (or class) in the image.

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

These metrics were especially important for evaluating performance on the **water class**, which was the primary focus of the project. The class index for water (usually set as 1) was used as the `focus_class` for focused reporting.

In addition to per-class metrics, the following aggregate metrics were computed across all classes: overall precision and recall, mean IoU (mIoU), overall dice.

All training experiments were logged using the **Weights & Biases (wandb)** platform.

The model performance is evaluated on Sentinel-2 and Landsat datasets. Additionally, the flood probability map is provided for flood Boris related samples.

6 Results

6.1 Experiments and Model Selection

Before settling on the final model architecture, a series of exploratory experiments were conducted, aimed at evaluating different segmentation models, loss functions, training configurations, and data augmentation strategies.

Several popular semantic segmentation models were evaluated based on their prevalence in relevant literature and their reported performance in satellite image segmentation tasks. The models tested included:

- U-Net + ResNet-50
- U-Net + ResNet-34
- U-Net + MobileNetV2

All models were trained on a smaller subset of the dataset ($\sim 3,000$ samples) for 4 epochs. These short runs were intended to provide a quick performance comparison under constrained training conditions. It was assumed that increasing data volume and training duration would improve performance, but early evaluations were useful for model elimination. Performance of all trained models can be seen in Figure 4)

Notably, DeepLabV3 models required larger image sizes to perform well due to their reliance on atrous convolutions and large receptive fields. When tested on 64×64 images, their performance was significantly worse than U-Net variants. Increasing the image resolution (e.g., to 128×128) yielded much better results for DeepLab, making it more competitive.

The code related to the experiments part can be found in the corresponding Python notebook.⁴

Loss Function Exploration

Initially, models were trained using only the cross-entropy loss. However, this led to suboptimal segmentation results—particularly in boundary regions and for small water bodies. To address this, a combined loss function was introduced: a weighted sum of cross-entropy and Dice loss, which better balances pixel-level classification with region-level consistency.

Further, there were experiments with different class weights in the cross-entropy loss to emphasize underrepresented classes such as water. While this improved recall for water in some cases, overly aggressive weighting led to overprediction, causing many false positives.

Data Augmentation Strategies

Various data augmentation strategies were evaluated to improve model generalization. The augmentations included geometric transforms (flips, rotations), intensity changes (brightness/contrast, gamma), and noise injection. However, most augmentation pipelines resulted in a decline in model performance. Simpler augmentations such as horizontal flips and mild brightness/contrast adjustments proved to be more reliable. No data augmentation techniques were used in the final pipeline.

Training Configuration Tuning

During early experiments, switching from a batch size of 64 to 128 led to more stable gradients and better convergence. This was particularly beneficial in improving class-wise segmentation accuracy.

The code related to the modelling results can be found in the corresponding Python notebook.⁵

6.2 Train & Validation Sets Performance

The results of the final model performance on train and validation sets are presented in the table and figures below. Additionally, Figure 4 shows metrics comparison during the training process for the final model and other model architectures considered. The focus is on metrics for class 1 as this serves as an indicator of how well waterbodies are segmented.

⁴[05_experiments.ipynb](#)

⁵[04_modelling.ipynb](#)



Figure 4: Performance of all considered models on train and validation sets.

6.3 Sentinel-2 Based Test Set Results

In this section, the results for evaluating model on the Sentinel-2 test set are discussed.

Table 1: Evaluation Metrics on the Test Set Sentinel-2 for Class 1 (Water)

Metric	Class 1 (Water)
Precision	0.839
Recall	0.898
IoU	0.766
Dice Coefficient	0.867

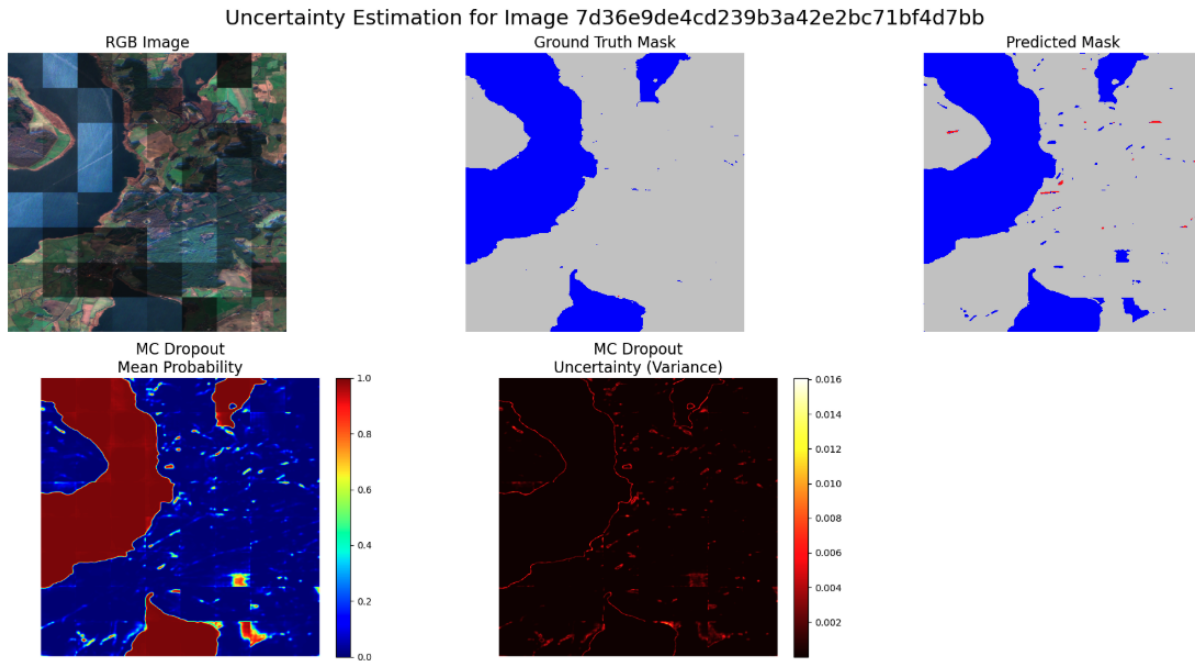


Figure 5: Visual example, showing patches of the same image, ground truth and predicted mask, uncertainty estimation as an output of MC Dropout procedure.

Quantitative evaluation on the test set demonstrates strong performance for water detection. As shown in Table 1, the model achieved a Dice coefficient of 0.867 and an IoU of 0.766 on the general Sentinel-2 test set. These results indicate a high degree of spatial overlap between predicted water regions and the ground truth, with both precision (0.839) and recall (0.898) values suggesting that the model is not only accurate but also robust in capturing a wide range of waterbody shapes and sizes.

These results directly address the research question Q2 - *How accurately can the proposed deep learning model identify water-covered areas in satellite imagery?*, showing that the proposed model is highly effective at identifying water-covered areas in Sentinel-2 imagery. The strong performance across multiple segmentation metrics confirms that the model generalizes well on unseen test data and is suitable for operational waterbody mapping tasks.

A visual example in Figure 5 further supports these findings. The model accurately delineates major water regions and preserves their overall shape and extent. The MC Dropout mean probability map shows strong confidence in core water areas (values close to 1), while the uncertainty map reveals elevated variance along water-land boundaries, highlighting regions where the model is less certain — typically around transitional or ambiguous zones. Additionally, the confidence map indicates that incorrectly classified areas outside the core water boundaries are

associated with high uncertainty, suggesting that prediction quality can be further improved by filtering out pixels below the confidence threshold.

6.4 Sentinel-2 Based Boris Flood Set Results

In this section, the results for evaluating model on the set of images made shortly after Boris storm are discussed. Images extracted from Sentinel-2 satellite.

Table 2: Evaluation Metrics on the Boris Set Sentinel for Class 1 (Water)

Metric	Class 1 (Water)
Precision	0.743
Recall	0.760
IoU	0.602
Dice Coefficient	0.751

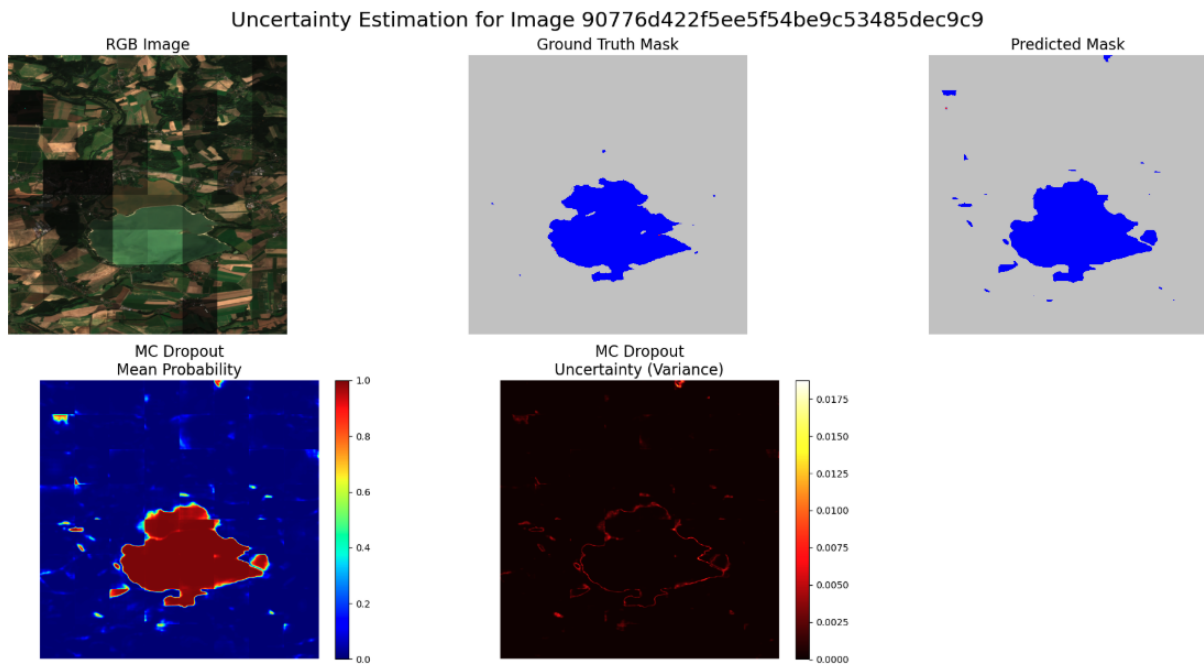


Figure 6: Visual example, showing patches of the same image, ground truth and predicted mask, uncertainty estimation as an output of MC Dropout procedure.

On the Boris flood Sentinel set (Table 2), performance metrics decrease slightly: Dice drops to 0.751 and IoU to 0.602. This decline is expected given the presence of more complex, dynamic water features during flooding, which often differ from the stable water patterns the model was primarily trained on. Nonetheless, the model maintains reasonable generalization to unseen conditions during flood, as shown in Figure 6.

6.5 Landsat Based Boris Flood Set Results

In this section, the results for evaluating model on the set of images made shortly after Boris storm are discussed. Images are extracted from Landsat satellite.

Table 3: Evaluation Metrics on the Boris Set Landsat for Class 1 (Water)

Metric	Class 1 (Water)
Precision	0.024
Recall	0.161
IoU	0.021
Dice Coefficient	0.042

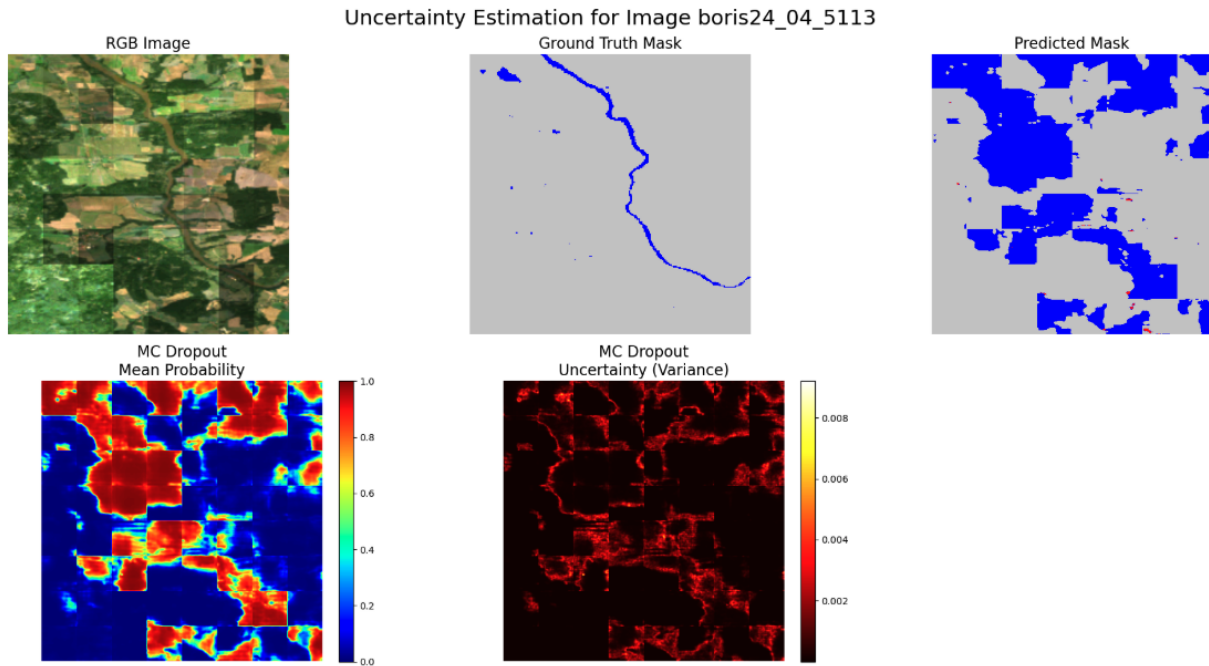


Figure 7: Visual example, showing patches of the same image, ground truth and predicted mask, uncertainty estimation as an output of MC Dropout procedure.

The model's performance on the Boris flood dataset using Landsat imagery was substantially lower, as shown in Table 3. The Dice coefficient dropped to 0.042 and the IoU to just 0.021, with precision at 0.024 and recall at 0.161. These results represent a significant degradation in segmentation quality compared to the Sentinel-2 test set.

This evaluation directly addresses the research question Q3 regarding model transferability: *To what extent can a model trained on Sentinel-2 data be transferred to Landsat imagery?* The sharp decline in all evaluation metrics indicates that cross-sensor generalization is highly limited in this case. The model, trained exclusively on Sentinel-2 data, struggles to interpret Landsat inputs, which differ in spatial resolution, spectral characteristics, and surface reflectance properties.

Figure 7 further illustrates this challenge. While the ground truth mask delineates river clearly, the predicted mask fails to capture water regions correctly, leading to a high false positive rate. The MC Dropout mean probability map shows high water probabilities across broad areas, including regions that are not water in the ground truth. This suggests that the model has learned to respond strongly to some features associated with water, but lacks the discriminative power to localize them accurately. The associated uncertainty map shows low to moderate uncertainty in the predicted regions, implying that the model is overconfident in

many of its incorrect predictions. This highlights the domain shift challenge and the need for improved domain adaptation or calibration strategies.

6.6 Flood Water Detection

In addition to training a model for water segmentation, a secondary objective of this project was to evaluate the model's performance in detecting water presence in **flooded areas**. Floodwaters are typically transient and spatially dynamic, in contrast to the relatively stable boundaries of lakes and rivers, which might result in a significant decrease in prediction quality.

To assess this, areas within the region of interest were examined shortly before and after flood events to identify newly emerged water surfaces. The following flooded areas were identified.

6.6.1 Lanžhot

A town located on the Czech-Slovak border, an area was flooded in the fields near this town. Flooded area can be seen in Figure 8, flooded areas are marked in grey, permanent waterbodies in blue.



(a) Before Flood, 08.09.2024



(b) After Flood, 23.09.2024

Figure 8: Comparison of the same region before and after flooding, NDWI index & permanent waterbodies

Figure 9 shows a visual example of model predictions for patches of a satellite image of exactly the same area captured during a flood event. The model successfully identifies the flooded areas with relatively high confidence. The RGB image, ground truth mask, and predicted mask are shown alongside outputs of Monte Carlo (MC) Dropout inference, specifically the mean probability map and uncertainty (variance) map. It can be seen that the MC Dropout Mean Probability map shows strong confidence (values close to 1.0) in most detected flood zones, the uncertainty (variance) map highlights boundaries and smaller regions with elevated uncertainty, which is consistent with edge areas or ambiguous zones.

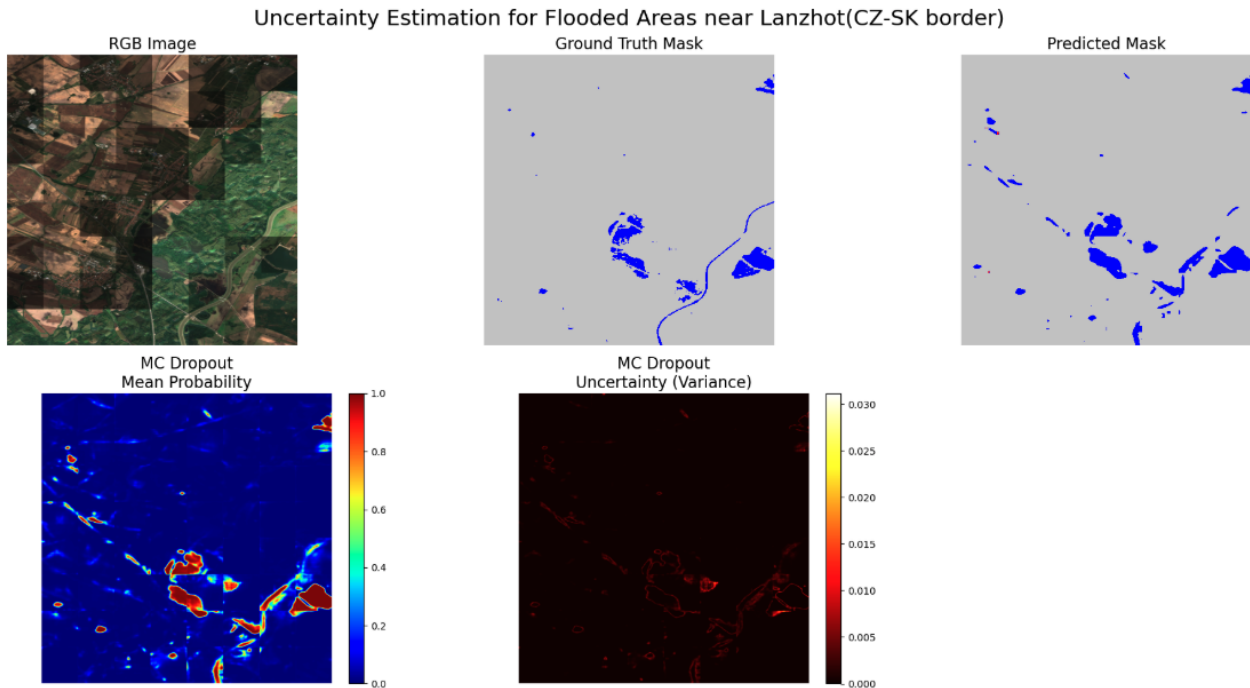


Figure 9: Visual example, showing patches of the same area, ground truth and predicted mask, uncertainty estimation as an output of MC Dropout procedure.

To further assess the reliability of predictions, performance metrics were analyzed across varying confidence thresholds, those thresholds were applied to the mean probability output from the MC Dropout approach. Predictions with confidence below a specified threshold were filtered out, under the assumption that low-confidence predictions may be less reliable.

The results of this threshold-based filtering are shown in Figure 10. The top chart illustrates how metrics such as Precision, Recall, F1 Score, and IoU vary with increasing confidence thresholds. The bottom row presents the visual effects of applying different thresholds to the predicted mask.

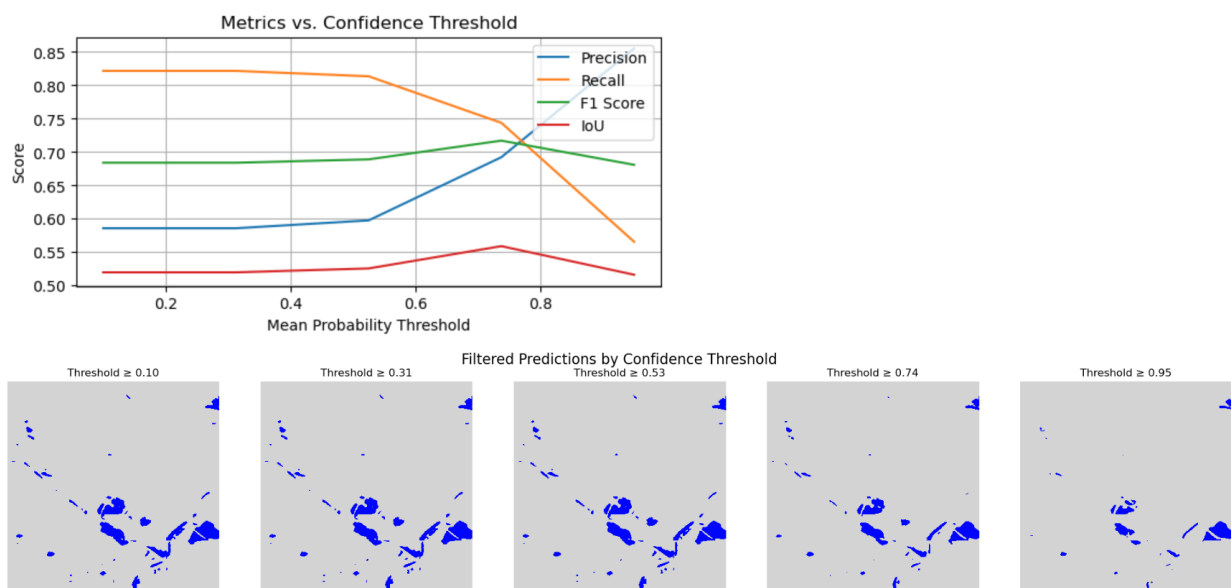


Figure 10: Quality metrics and predicted masks at different MC Dropout confidence thresholds.

It can be seen that as the mean probability threshold increases, precision improves—indicating more reliable positive predictions—while recall decreases, as fewer predicted pixels are retained. The F1 Score and IoU reach their peak at a moderate threshold (approximately 0.53–0.74), reflecting a balance between coverage and reliability. Visual inspection of the predictions reveals that the model tends to overpredict water pixels in this image; therefore, filtering out low-confidence pixels can improve the F1 score, which can be seen in the metrics chart. This demonstrates that MC Dropout outputs can enhance the quality of final predictions, though the confidence threshold should not be set too strictly to avoid missing true positives.

6.6.2 Kisoroszi

A village located in Hungary between 2 branches of Danube river, an area was flooded close to this village. Flooded area can be seen in Figure 11, flooded areas are marked in grey, permanent waterbodies in blue.

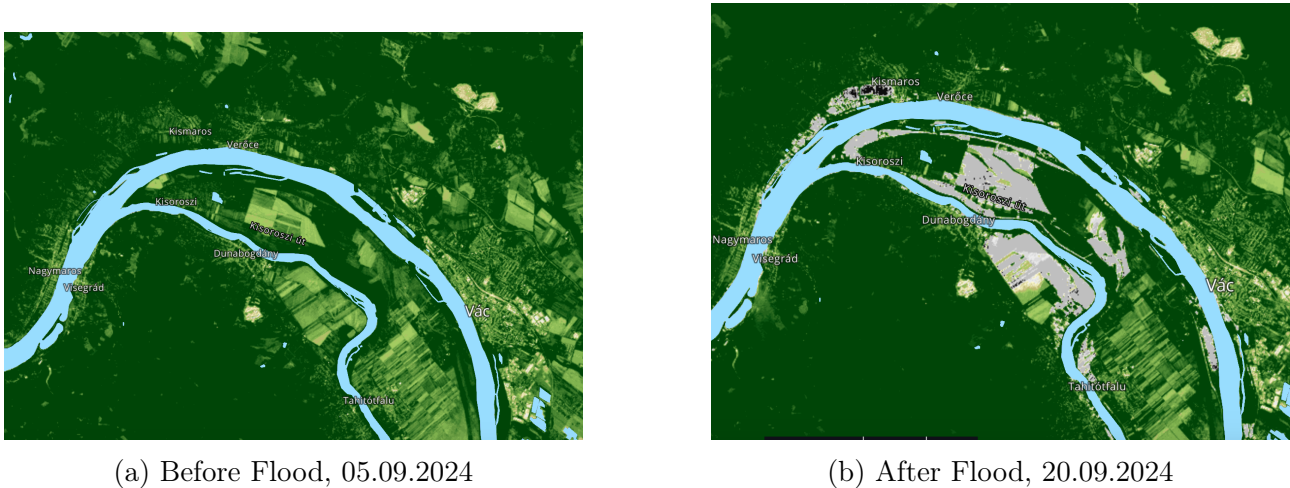


Figure 11: Comparison of the same region before and after flooding, NDWI index & permanent waterbodies.

To further validate the model’s generalization ability, another area was examined. The flood water, in contrast to the previous case, is clearly visible; the image is shown in Figure 12. Corresponding ground truth, model prediction, and MC Dropout-based uncertainty outputs are also presented in the same figure.

For this example, the model is able to identify several flooded regions; however, under-segmentation is noticeable, particularly in smaller or less visually distinct water patches. The mean probability map obtained from the MC Dropout approach shows that the model assigns higher confidence to larger, more pronounced flooded areas, while boundaries and fragmented regions are associated with lower confidence levels.

One possible reason for this degraded performance is the visual variability of water surfaces during flood events. Specifically, water appearance may change significantly in terms of colour, texture, and reflectance due to rainfall or vegetation debris. These changes make flooded water surfaces less similar to the stable and clearer waterbodies (e.g., lakes and rivers under normal conditions) that the model was predominantly trained on. As a result, the model may struggle to generalize to flood conditions.

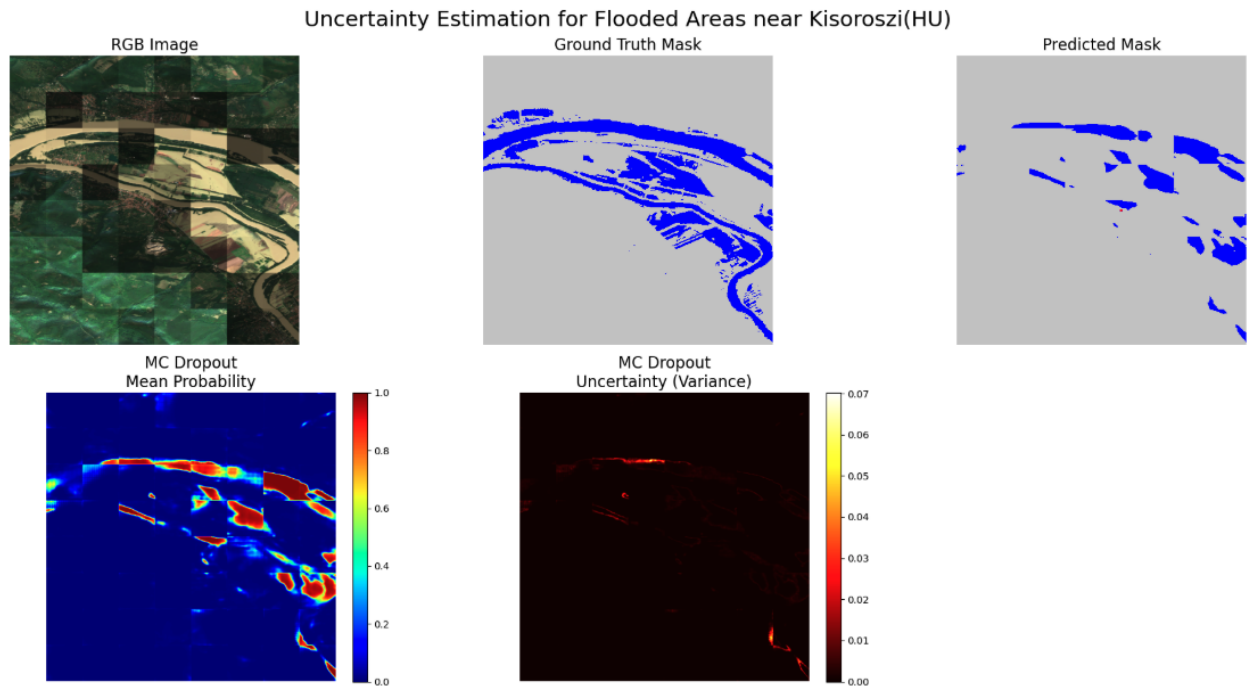


Figure 12: Visual example, showing patches of the same area, ground truth and predicted mask, uncertainty estimation as an output of MC Dropout procedure.

In Figure 13, the behaviour of prediction quality across multiple MC Dropout confidence thresholds was analyzed. Similar to the previous example, precision increases with higher thresholds, indicating more reliable water predictions when only confident pixels are retained. Recall, being not very high from the beginning, drops significantly beyond a threshold of approximately 0.5, as the model begins to discard more low-confidence (but potentially correct) water predictions. Both F1 Score and IoU peak around intermediate thresholds (approximately 0.31–0.53), reflecting a trade-off between capturing all flood areas and avoiding false positives.

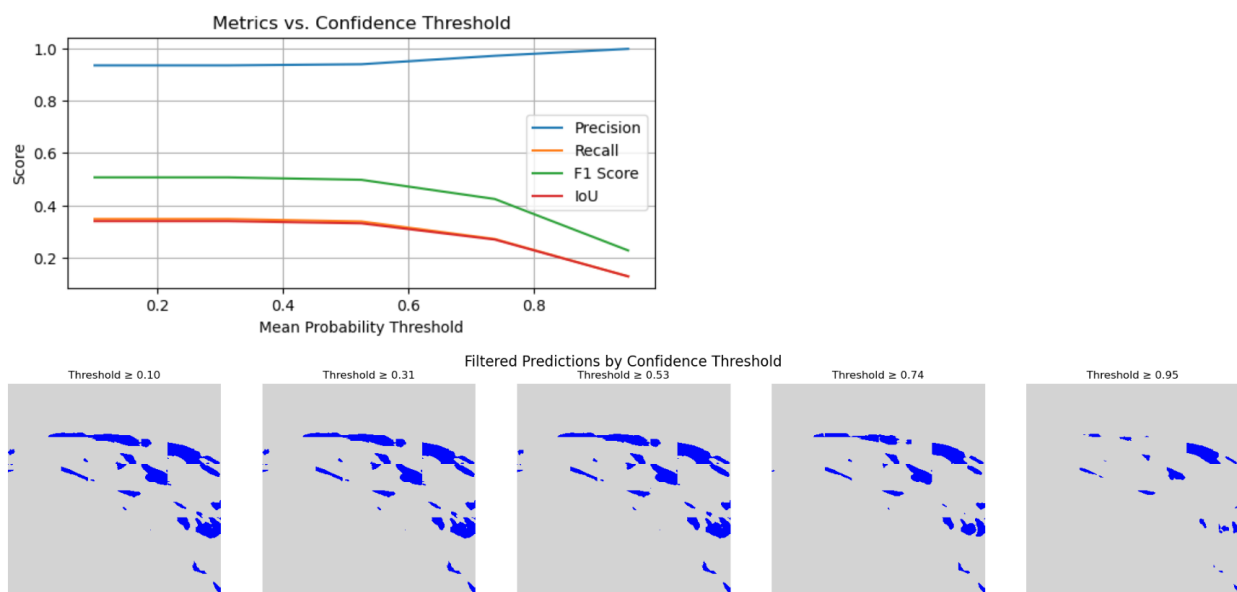


Figure 13: Quality metrics and predicted masks at different MC Dropout confidence thresholds.

After analyzing two cases where the presence of floodwater was evident, it can be concluded that the model was able to identify some flooded areas. However, the quality metrics for both images were lower than those obtained on the test set. This indicates a limitation arising from the model being trained primarily on water bodies under normal conditions, which affects its ability to accurately detect different types of flooding.

This part of the project directly relates to the final research question — Q4 — *Can uncertainty estimates be used to improve the reliability of flood classification?* Although the approach of calculating quality metrics across different confidence thresholds was tested on only two cases, it proved useful in addressing both model overconfidence (Case 1) and underconfidence (Case 2). Therefore, it can be concluded that confidence maps are valuable tools; however, the threshold should not be set too strictly. Instead, it should be selected based on case-by-case evaluation.

7 Conclusion

This project presented a pipeline for detecting water bodies and potential flood regions, primarily using Sentinel-2 satellite imagery, while also evaluating the method's applicability to Landsat data.

The DeepLabV3 architecture with a ResNet-34 backbone was chosen after comparing several alternatives, including U-Net variants and lightweight encoders. The final model demonstrated strong performance on the Sentinel-2 test set and generalized reasonably well to a separate test set comprising images captured shortly after the Boris storm event. However, performance declined significantly when the model was applied to Landsat imagery.

Finally, the model was evaluated on two real-world flood cases, each representing different visual characteristics of floodwater. Unfortunately, the model's performance was suboptimal: it tended to overpredict in one case and underpredict in the other. However, by incorporating uncertainty maps generated through the Monte Carlo dropout procedure, the precision-recall trade-off could be better managed, leading to improvements in overall quality metrics such as the F1-score and Intersection over Union (IoU).

The following answers can be given to the research questions stated at the beginning of the project:

1. **Q1: To what extent can high-quality waterbody masks be generated using NDWI and Sentinel-2 Scene Classification Layer (SCL) data**

For the Sentinel-2 part of the dataset, approximately 24% of images required manual corrections, indicating that around three-quarters of the auto-generated masks were of sufficient quality for direct use in training. These findings suggest that NDWI and SCL-based methods offer a viable starting point for semi-automated label generation in waterbody segmentation tasks, significantly reducing manual annotation effort.

2. **Q2: How accurately can the proposed deep learning model identify water-covered areas in satellite imagery?**

Quantitative evaluation on the Sentinel-2 test set of the proposed model architecture demonstrates strong performance for water detection. The model achieved a Dice coefficient of 0.867 and an IoU of 0.766. These results indicate a high degree of spatial overlap between predicted water regions and the ground truth with balanced precision (0.839)

- recall (0.898) relationship, meaning that the proposed model is effective at identifying water-covered areas in Sentinel-2 imagery.

3. Q3: To what extent can a model trained on Sentinel-2 data be transferred to Landsat imagery?

The model's performance on the Boris flood dataset using Landsat imagery was substantially lower, the Dice coefficient dropped to 0.042 and the IoU to just 0.021, with precision at 0.024 and recall at 0.161. These results represent a significant degradation in segmentation quality compared to the Sentinel-2 test set, meaning that the model, trained exclusively on Sentinel-2 data, struggles to interpret Landsat inputs, which differ in spatial resolution, spectral characteristics, and surface reflectance properties.

4. Q4: Can uncertainty estimates be used to improve the reliability of flood classification?

Although the approach of calculating quality metrics across different confidence thresholds was tested on only two cases, it proved to be useful in addressing both model overconfidence (Case 1) and underconfidence (Case 2). Therefore, it can be concluded that confidence maps are valuable tools; however, the threshold should not be set too strictly. Instead, it should be selected based on case-by-case evaluation.

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